

ECON 312 — Intermediate Microeconomics

Lecture Notes — Ch. 4: Utility

Fall 2026

Before: assign Khan Academy on partial derivatives.

Utility: a ① way of ② assigning a number to every possible consumption bundle such that ③ **more-preferred** bundles get larger numbers than less preferred bundles.

i.e. if $(x_1, x_2) \succsim_{\text{preference}} (y_1, y_2)$, then $u(x_1, x_2) \geq u(y_1, y_2)$.

u represents $\succsim$

Outline

- ① (A Way) — **Utility functions aren't unique**
- ② (Assigning) How to build a utility function
- ③ (More vs. Less) Why utility representations exist
- ④ $\Rightarrow$ Why even care...

a major disclaimer

1. A Way

Suppose $A \succ B \succ C$. Here are some utility options:

Bundle	U_1	U_2	U_3	U_4	U_5	U_6
A	3	1,000,000	-3	4	π	∞
B	2	999,999	-10	7	$\pi/2$	5
C	1	-2	-18	-1	0	-2
Q Utility representation?	✓	✓	✓	✗	✓	✗

What matters? The order, not the size (cardinality).

$\rightarrow$ How would *cardinal* preferences work? Would $u(x) = 2, u(y) = 4$ mean that I would pay $2x$ for y ? $\rightarrow$ what if \$ is in y ...

* u represents $\succsim$ but is not $\succsim$. Many different u 's can represent $\succsim$.

If u and v both represent $\succsim$, then there is a **monotonic transformation** f such that

$$u(x) = f(v(x)).$$

Def: f is a **monotonic function** if for every $x > y$, then $f(x) > f(y)$.

$f(v)$ is a monotonic transformation of v if f is a monotone function.

(the difference is just semantics)

One way to check if a (differentiable) function is monotonic is to check the derivative:

$$f'(u) = \lim_{\hat{u} \rightarrow u} \frac{f(\hat{u}) - f(u)}{\hat{u} - u} \geq 0 \quad \text{b/c } f(\hat{u}) - f(u) \text{ and } \hat{u} - u \text{ always have the same sign.}$$

Theorem

If u represents $\succsim$, then **any** monotonic transformation of u also represents $\succsim$.

Interpretation: $\succsim$ just tells us the order on bundles and nothing else. u is just a name of $\succsim$, and the numeric value only matters *through* $\succsim$, nothing else.

→ I can add a million, multiply by π , cube it, what have you. u is *only* a name!

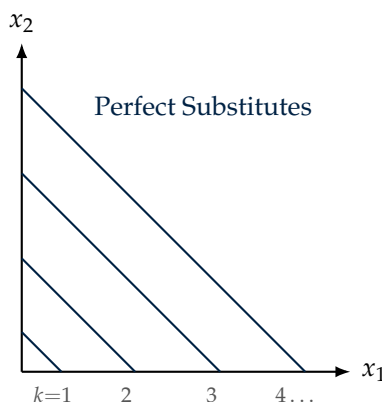
2. Assigning

How do we actually write/find a utility function?

Two ways

- ①a Find indifference curves.
- ①b Find a function with the same level sets —

→ all (x_1, x_2) s.t. $u(x_1, x_2) = u(y_1, y_2)$, e.g. $x_1 + x_2 = k$



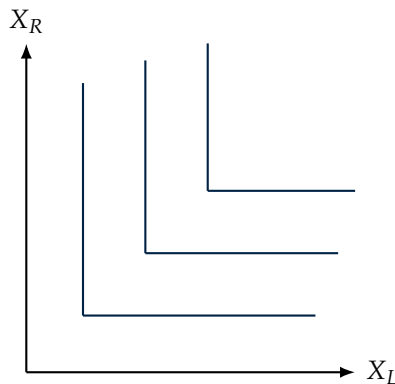
→ easy here b/c preferences line up nicely w/ utility...

- ②a Think about what preferences “represent” and how the consumer is trying to maximize them.
 ②b Then guess...

→ *More art than science; so let's practice!*

Example 1: Perfect Complements

Left shoes and right shoes.

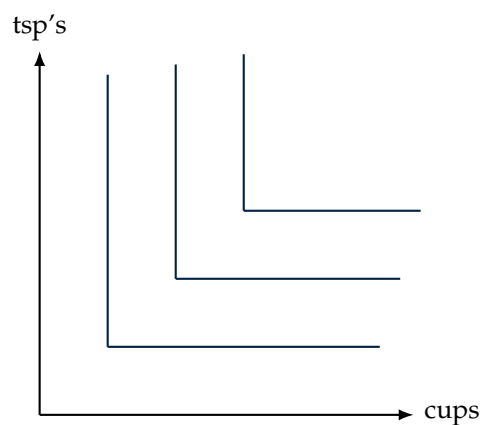


- (a) Draw indifference curves.
 (b) They really care about which has a *lower* value... → take the minimum of x_1 & x_2 .

$$u(x_1, x_2) = \min\{x_1, x_2\} \quad \checkmark$$

But what if instead it is cups of tea + sugar?

2 tsp's of sugar per cup of tea



□ Should it be $u(x_{tea}, x_{sugar}) = \min\{x_{tea}, x_{sugar}\}$?

No! →

$u(5, 5) = 5$ but I only drink 2.5 cups

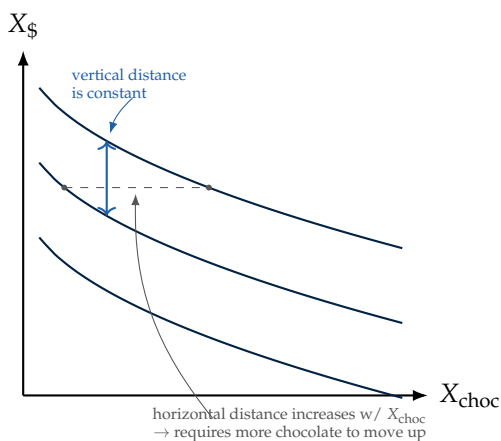
$u(4, 8) = 4$ but I drink 4 cups!

Hmm — so we need to multiply:

$$u(x_{\text{tea}}, x_{\text{sugar}}) = \min \left\{ x_{\text{tea}}, \frac{1}{2} x_{\text{sugar}} \right\} \quad \uparrow \text{b/c } 2 \text{ sugars} = 1 \text{ tea} \dots$$

Example 2: A Good + Other Income

Chocolate + all other consumption.



Q How should we think about preferences for chocolate?

Are you willing to pay as much for the 2nd bar as for the first?

(a) What do these indifference curves look like?

(b) What are they maxing?

→ $X_{\$}$ is just raw utility.

→ X_{choc} has a decreasing derivative. Why not $\sqrt{X_{\text{choc}}}$?

[any concave function works]

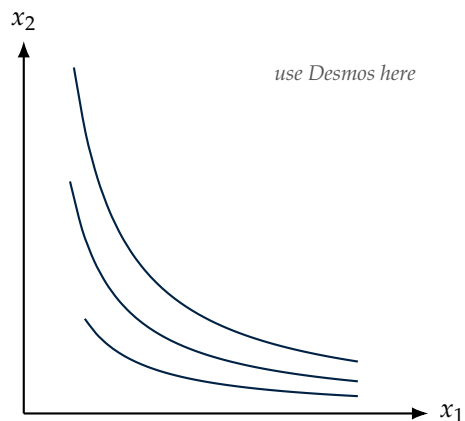
$$u(x_{\text{choc}}, x_{\$}) = x_{\$} + \sqrt{x_{\text{choc}}} \quad \checkmark$$

Bonus Example: Cobb–Douglas

This one doesn't have a simple preference story, but it captures the idea of *imperfect complements*. Here is the utility function:

$$u(x_1, x_2) = x_1^c x_2^d \quad \text{w/ } c, d \geq 0.$$

It looks like:



Two useful tricks:

1. Raise the utility to the power of $\frac{1}{c+d}$:

$$u(x_1, x_2) = x_1^{\frac{c}{c+d}} x_2^{\frac{d}{c+d}} \quad \leftarrow \text{is a monotonic transformation}$$

Set $a = \frac{c}{c+d}$, and get $u(x_1, x_2) = x_1^a x_2^{1-a}$.

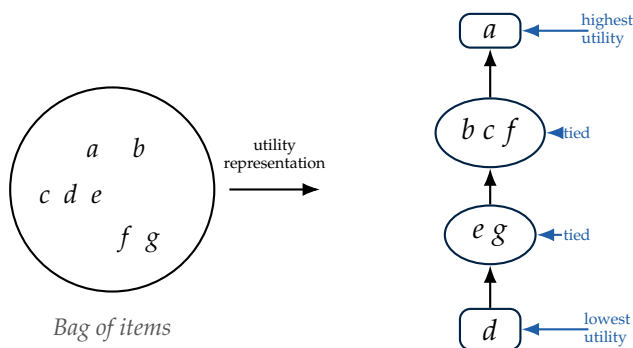
2. Take a natural log:

$$u(x_1, x_2) = a \ln(x_1) + (1-a) \ln(x_2).$$

3. More vs. Less

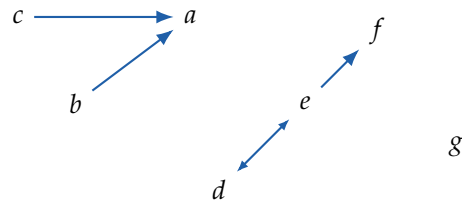
Why does a utility representation exist?

→ First, let's visualize a utility representation.



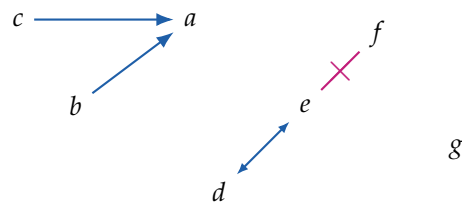
(stacked with an arrow → strictly preferred; grouped in an oval → tied)

But allowing $\succsim$ to be *anything* means we could get:



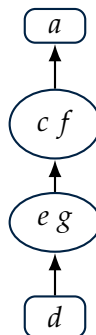
... which is nuts.

(A) Require Transitivity



(the missing $e \rightarrow f$ link is exactly what transitivity would force)

(B) Require Completeness



Side note: when we go to an infinite number of bundles, we need an additional technical axiom (continuity).

4. Why Even Care

Before answering this, we need to relate the **Marginal Rate of Substitution** to the **Marginal Utility**.

→ $\frac{dx_2}{dx_1}$ of the indifference curve (of x_1)

Marginal = derivative

Marginal Utility: how much utility the consumer gains from a teeny-tiny increase in (x_1).

$$MU_1 = \lim_{\Delta x_1 \rightarrow 0} \frac{u(x_1 + \Delta x_1, x_2) - u(x_1, x_2)}{\Delta x_1} = \frac{\partial u(x_1, x_2)}{\partial x_1}$$

This is the partial derivative: I'm just changing x_1 while leaving x_2 constant!

Change in Utility = Marginal Utility $\times$ size of increase

$$dU = MU_1 \cdot dx_1 \quad \checkmark \quad \Rightarrow \quad dU = MU_2 \cdot dx_2 \quad \checkmark$$

Pause here for just a second:

MU_1 scales w/ u ... if you sub $v = 2u$, then you get $2MU_1$ as well...

$\Rightarrow MU_1$ has no behavioral content $\Rightarrow$ I can't tell its magnitude from any data...

... Now back to the program.

Let's now use our new notation to write the MRS.

$\Rightarrow$ 2 methods

Method 1: Differentials

Idea: keep U unchanged by adding/subtracting dx_1 and dx_2 .

$$\underbrace{\frac{dU}{\text{change in } u}} = \underbrace{\frac{dU}{\text{change from } x_1}} + \underbrace{\frac{dU}{\text{change from } x_2}} = \underbrace{0}_{\text{no change}}$$

$$MU_1 dx_1 + MU_2 dx_2 = 0$$

$\Rightarrow$ Solve for MRS = $\frac{dx_2}{dx_1}$:

$$\frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2} = -\frac{\partial u(x_1, x_2)/\partial x_1}{\partial u(x_1, x_2)/\partial x_2} \quad \checkmark$$

Do a Cobb–Douglas example here.

Method 2: Implicit Function

Idea: map out the “indifference curve”; take the derivative on this curve.

Set $x_2(x_1)$ [a function] so that $u(x_1, x_2(x_1)) = k$

picks the x_2 to keep utility constant

Total. Take the derivative w/ respect to x_1 :

$$\underbrace{\frac{d}{dx_1} [u(x_1, x_2(x_1))]}_{\text{total: how does changing the variable } x_1 \text{ affect things}} = \underbrace{\frac{\partial}{\partial x_1} u(x_1, x_2)}_{\text{partial: how does changing the 1st argument matter?}} + \frac{\partial}{\partial x_2} u(x_1, x_2) \cdot \underbrace{\frac{\partial x_2(x_1)}{\partial x_1}}_{\text{chain rule: change the 2nd argument at the rate caused by change in first}}$$

Left-hand side: $\frac{d}{dx_1} k = 0$, so

$$\frac{\partial}{\partial x_1} u(x_1, x_2) + \frac{\partial}{\partial x_2} u(x_1, x_2) \cdot \frac{\partial x_2(x_1)}{\partial x_1} = 0$$

Solve:

$$\frac{\partial x_2(x_1)}{\partial x_1} = -\frac{\partial u(x_1, x_2)/\partial x_1}{\partial u(x_1, x_2)/\partial x_2} \quad \checkmark$$

Same as before.

Back to “No Behavioral Content”

Look at $MRS = -\frac{MU_1}{MU_2}$.

If I scale u by 2, look what happens:

$$MRS = -\frac{2MU_1}{2MU_2} = -\frac{MU_1}{MU_2} \dots$$

So the scaling falls out!

Formally, let $v(x_1, x_2) = f(u(x_1, x_2))$ for a monotonic f . Then, by the **chain rule**,

$$MRS = -\frac{\partial v / \partial x_1}{\partial v / \partial x_2} = -\frac{\cancel{\partial f / \partial u} \cdot \partial u / \partial x_1}{\cancel{\partial f / \partial u} \cdot \partial u / \partial x_2} = -\frac{MU_1}{MU_2}$$

⇒ So the MRS (which has behavioral content) nails down the ratio of the marginal utilities!

Preferences let us compare things, but Marginal Utility gives us a way to estimate **relative** importance.

Example: Utility for an EV

Assume a utility function with the form

$$u(\underbrace{\text{FC}}_{\text{fuel cost saving}}, \underbrace{\text{ER}}_{\text{emission reduction}}, \underbrace{\text{DR}}_{\text{driving range}}, \underbrace{\text{CT}}_{\text{charge time}}, \underbrace{\text{PP}}_{\text{purchasing price}}) \\ = 0.21 \text{ FC} + 0.10 \text{ ER} + 0.49 \text{ DR} \\ + 0.19 \text{ CT} - 0.09 \text{ PP}$$

The weights themselves are meaningless, but the **ratios** matter!

→ You need twice as much ER to match a gain from FC!